SQL Analysis

Total Number of Transactions

A query was executed to determine the total number of transactions present in the dataset.

Query:

```
SELECT COUNT(*) AS total_transactions
FROM cc_data;
```

Insights:

total_transactio...
389002

- The dataset contains a large number of transactions, indicating sufficient data for analysis.
- A higher number of records improves the reliability of insights and fraud detection patterns.
- This value serves as a baseline for calculating fraud percentages and other metrics.

Top 10 Most Frequent Merchants

This analysis identifies the merchants with the highest number of transactions.

Query:

```
SELECT merchant, COUNT(*) AS transaction_count
FROM cc_data
GROUP BY merchant
ORDER BY transaction_count DESC
LIMIT 10;
```

Insights:

merchant	transaction_co...
fraud_Kilback LLC	1305
fraud_Cormier LLC	1120
fraud_Kuhn LLC	1058
fraud_Schumm PLC	1024
fraud_Dickinson Ltd	997
fraud_Boyer PLC	990
fraud_Jenkins, Hauck and Friesen	856
fraud_Rodriguez Group	839
fraud_Eichmann, Bogan and Rodriguez	829
fraud_Christiansen, Goyette and Schamberger	828

- A small number of merchants account for a large share of transactions.
- High-frequency merchants may represent popular platforms or services.
- Monitoring these merchants is important, as high volume can also attract fraudulent activity.

Average Transaction Amount by Category

This query calculates the average transaction amount for each category.

Query:

```
SELECT category, AVG(amt) AS avg_transaction_amount
FROM cc_data
GROUP BY category;
```

Insights:

category	avg_transaction_amo...
misc_net	82.03714838844692
grocery_pos	117.5270554633738
shopping_net	88.52703420495718
personal_care	48.12700690461437
gas_transport	63.39943986072221
entertainment	64.09267228504379
home	58.292295579587204
food_dining	51.19593018101227
kids_pets	57.602997316188
travel	114.35104240137805
misc_pos	61.842209331337294

- Transaction amounts vary significantly across categories.

- Some categories show higher average spending, indicating potential high-value transaction segments.
- Categories with unusually high averages may require closer monitoring for fraud.

Fraudulent Transactions and Percentage

This analysis calculates the number of fraudulent transactions and their percentage of total transactions.

Query:

```
SELECT
    SUM(is_fraud) AS fraud_count,
    COUNT(*) AS total_transactions,
    (SUM(is_fraud) * 100.0 / COUNT(*)) AS fraud_percentage
FROM cc_data;
```

Insights:

fraud_count	total_transactio...	fraud_percenta...
2252	389002	0.57892

- Fraudulent transactions represent a small percentage of the total dataset, indicating class imbalance.
- Even a small percentage can result in significant financial loss.
- This highlights the importance of accurate fraud detection systems.

Joining Transaction and Location Data

A join operation was performed to combine transaction data with location data to obtain latitude and longitude for each transaction.

Query:

```
SELECT
    c.*,
    l.lat,
    l.long
FROM cc_data c
JOIN location_data l
ON c.cc_num = l.cc_num;
```

Insights:

- The joined dataset enables geographical analysis of transactions.
- Location-based patterns can help identify unusual or suspicious activity.
- This is useful for detecting fraud occurring in unexpected locations.

City with Highest Population

This query identifies the city with the highest population in the dataset.

Query:

```
SELECT city, city_pop
FROM cc_data
ORDER BY city_pop DESC
LIMIT 1;
```

Insights:

city	city_pop
Houston	2906700

- The city with the highest population is likely to have a higher number of transactions.
- Larger cities generally contribute more to transaction volume.
- Population size can influence transaction behavior and risk exposure.

Earliest and Latest Transaction Dates

This analysis determines the time range of the dataset.

Query:

```
SELECT
    MIN(trans_date_trans_time) AS earliest_date,
    MAX(trans_date_trans_time) AS latest_date
FROM cc_data;
```

Insights:

earliest_date	latest_date
2019-01-01	2020-06-21

- The dataset spans across a defined time period, enabling time-based analysis.
- Identifying the time range is important for trend analysis and seasonality detection.
- It also helps in understanding how recent the data is.

Total Amount Spent Across All Transactions

This query calculates the total value of all transactions.

Query:

```
SELECT SUM(amt) AS total_amount_spent
FROM cc_data;
```

Insights:

total_amount_spent
27402136.520000726

- The total transaction amount reflects the scale of financial activity in the dataset.
- Higher total value increases the potential impact of fraudulent activities.
- This metric is useful for business-level analysis.

Number of Transactions per Category

This analysis counts how many transactions occur in each category.

Query:

```
SELECT category, COUNT(*) AS transaction_count
FROM cc_data
GROUP BY category;
```

Insights:

category	transaction_co...
misc_net	19112
grocery_pos	36763
shopping_net	29294
personal_care	27373
gas_transport	39633
entertainment	28122
home	36897
food_dining	27070
kids_pets	33907

- Some categories dominate transaction volume.
- High-frequency categories indicate common spending patterns.
- Fraud detection models should pay special attention to high-volume categories.

Average Transaction Amount by Gender

This query compares average transaction amounts between genders.

Query:

```
SELECT gender, AVG(amt) AS avg_transaction
FROM cc_data
GROUP BY gender;
```

Insights:

gender	avg_transaction
F	69.85478898471565
M	71.15468427669961

- There may be slight differences in spending behavior across genders.
- However, gender alone is not a strong indicator of fraud.
- It is more useful when combined with other features.

Day of the Week with Highest Average Transaction Amount

This analysis identifies which day has the highest average transaction amount.

Query:

```
SELECT
    DAYNAME(trans_date_trans_time) AS day_of_week,
    AVG(amt) AS avg_transaction
FROM cc_data
GROUP BY day_of_week
ORDER BY avg_transaction DESC
```

Insights:

day_of_week	avg_transaction
Friday	71.56510343775214
Monday	71.3411641939844
Wednesday	70.61137554750991
Thursday	70.52343548895196
Saturday	70.25117172928354
Tuesday	69.63631613119342
Sunday	69.37245227039891

- Certain days show higher average spending, possibly due to lifestyle or behavioral patterns.
- Weekends or specific days may have higher transaction values.
- Understanding these patterns helps distinguish normal vs abnormal behavior.

Conclusion (SQL Phase)

The SQL analysis provided structured insights into transaction patterns, fraud distribution, and key behavioral trends. By aggregating and querying the dataset, important factors such as high-risk categories, transaction volumes, and temporal patterns were identified. These insights form a strong foundation for further exploratory data analysis and machine learning modeling.